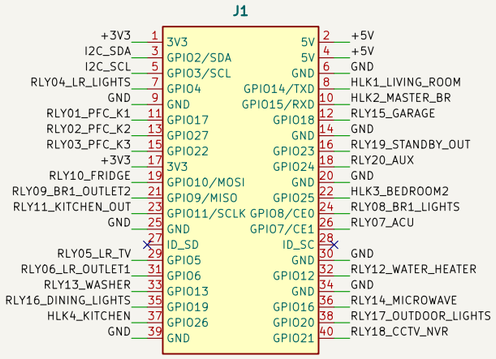
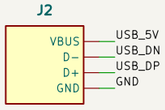
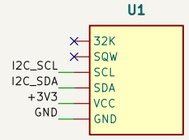
RASPBERRY PI 5 DOMAIN (relays, presence sensors, RTC)



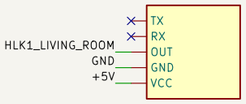
Raspberry Pi 5 — 40-pin GPIO header



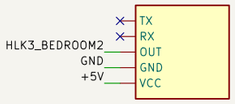
Raspberry Pi 5 — USB-A port



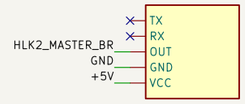
DS3231 RTC module (I2C 0x68)



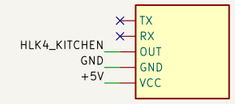
HLK-LD2410B zone1 — Living Room



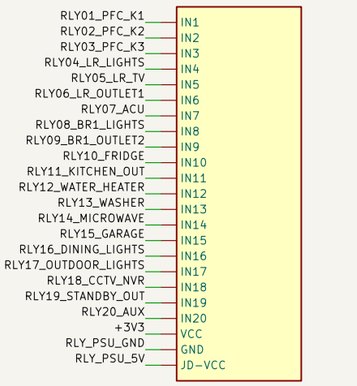
HLK-LD2410B zone3 — Bedroom2



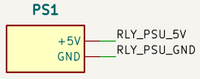
HLK-LD2410B zone2 — Master BR



HLK-LD2410B zone4 — Kitchen



20-ch opto relay board (active-LOW)



5V >=2A relay coil supply

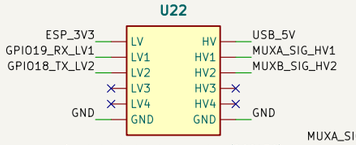
WIRING NOTES
1. Same label name = same wire. Connect every pin that shares a label.
   Full pin-to-pin list: NETLIST.md
2. J1 pin numbers are PHYSICAL header pins (not BCM).
3. K1 relay board: REMOVE the VCC/JD-VCC jumper.
   VCC = Pi 3V3 (opto LEDs); JD-VCC + GND = PS1 (5V >=2A).
   Do NOT tie Pi GND to K1 GND (keeps opto isolation).
4. Relays are active-LOW: GPIO LOW = relay ON.
   RLY01-03 drive the PFC capacitor-bank contactors.
5. GPIO14/15 (HLK zones 1-2) are UART pins:
   disable serial console + serial hardware.
6. Keep SPI (GPIO7-11) and audio/PWM overlays
   (GPIO12/13/18/19) disabled.
7. Boot check: GPIO9-27 pull down at power-up, so relays
   may click ON until app.py starts. Fix if needed:
   config.txt: gpio=9=13,16=24,27=op,dh
8. PZEM TTL '5V' pins, both mux VCC and the IR modules
   run on 5V (ESP32 VIN = USB_5V). The ESP32 is 3.3V, so
   RX/TX pass through the level converter U22 (LV1/LV2).
9. Two muxes, same S0-S3 wires: Mux A sends the selected
   PZEM TX to ESP32 RX1, Mux B sends ESP32 TX1 to that
   PZEM RX only. Channel Cn = PZEM #n on BOTH muxes.
10. Both mux EN = GND (always on). C11-C15 unused.
11. Control wiring only: mains side not shown.

MUX WIRING (2x CD74HC4067, Mux A + Mux B)

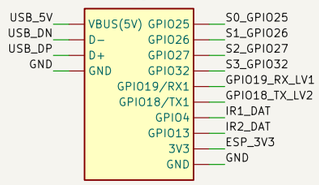
Mux pin	Mux A (U7)	Mux B (U19)
S0	ESP32 GPIO25	ESP32 GPIO25 (same wire)
S1	ESP32 GPIO26	ESP32 GPIO26 (same wire)
S2	ESP32 GPIO27	ESP32 GPIO27 (same wire)
S3	ESP32 GPIO32	ESP32 GPIO32 (same wire)
SIG	U22 HV1 (GPIO19 RX)	U22 HV2 (GPIO18 TX)
VCC	ESP32 5V (VIN)	ESP32 5V (VIN)
GND	ESP32 GND	ESP32 GND
EN	GND	GND
C0-C10	TX of PZEM #0-#10	RX of PZEM #0-#10
C11-C15	open	open

- S0-S3, VCC and GND go to BOTH muxes: one wire from the ESP32,
  then jump Mux A S0 -> Mux B S0 (or split it with a Y splice).
  Same for S1, S2, S3, VCC, GND/EN. Only SIG is different.
- Channel Cn = PZEM #n on BOTH muxes: its TX on Mux A Cn,
  its RX on Mux B Cn. Wrong channel = that PZEM shows STANDBY.
- Both muxes switch together: ESP32 asks PZEM #n through Mux B,
  PZEM #n answers through Mux A.

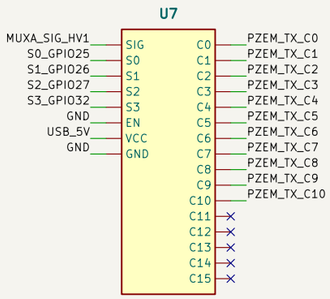
ESP32 DOMAIN (11x PZEM-004T via 2x CD74HC4067 at 5V, level converter, 2x aircon IR)



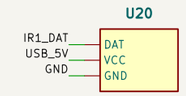
4-ch logic level converter (BSS138)



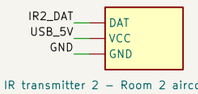
ESP32 DevKit (USB to Pi)



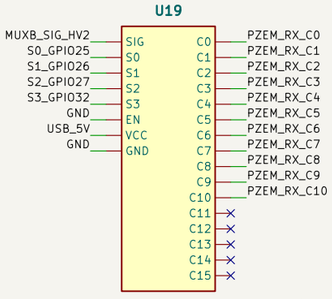
Mux A CD74HC4067 (PZEM TX -> ESP32 RX)



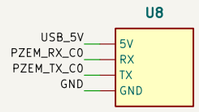
IR transmitter 1 — Room 1 aircon (AUX)



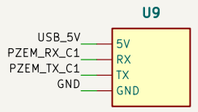
IR transmitter 2 — Room 2 aircon (Panasonic)



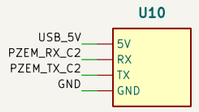
Mux B CD74HC4067 (ESP32 TX -> PZEM RX)



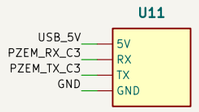
PZEM #0 Main Panel



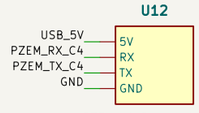
PZEM #1 Branch 1 (R4)



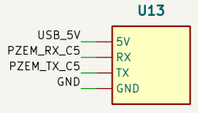
PZEM #2 Branch 2 (R5)



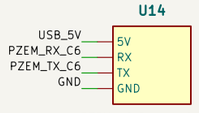
PZEM #3 Branch 3 (R6)



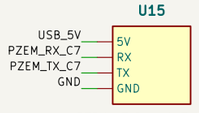
PZEM #4 Branch 4 (R7)



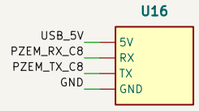
PZEM #5 Branch 5 (R8)



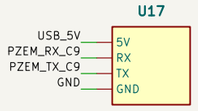
PZEM #6 Branch 6 (R9)



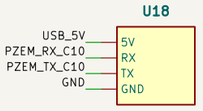
PZEM #7 Branch 7 (R10)



PZEM #8 Branch 8 (R11)



PZEM #9 Branch 9 (R12)



PZEM #10 Branch 10 (R13)

LEVEL CONVERTER U22 (4-ch, LV = 3.3V side, HV = 5V side)

U22 pin	Connects to	U22 pin	Connects to
LV	ESP32 3V3	HV	ESP32 5V (VIN)
LV1	ESP32 GPIO19 (RX)	HV1	Mux A SIG
LV2	ESP32 GPIO18 (TX)	HV2	Mux B SIG
LV3, LV4	unused	HV3, HV4	unused
GND	GND	GND	GND

- Only RX and TX need it: they carry the PZEM's 5V signals.
- S0-S3 go straight from the ESP32 (3.3V is read as HIGH by the
  74HC4067; it switches near half of VCC). If a PZEM ever reads
  the wrong channel, add a 2nd converter on S0-S3.

Control (low-voltage) wiring only
Generated by generate_schematic.py from app.py + ESP32 .ino
REMS / RECMS Thesis

Sheet: /
File: rems-control-wiring.kicad_sch

Title: REMS — Control Wiring Schematic

Size: A3	Date: 2026-10-30	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 1/1